

Deep Learning vs. Conventional Methods for Tabular Data Imputation

immediate

Abstract

The abstract should be a single paragraph written in plain language that a general reader can understand. Do not include citations, figures, tables, or undefined abbreviations in the abstract. Any abbreviations that appear in the title should be defined in the abstract. The length should not exceed 250 words, to include:

- An opening sentence that states the question/problem addressed by the research AND
- Enough background content to give context to the study AND
- A brief statement of primary results AND
- A short concluding sentence.

Keep in mind that abstracts are the primary way research is discovered in databases. Insert keywords naturally into the abstract text, using terms that a reader might actually type into a search bar.

1 Introduction

Your manuscript should contain all of the numbered sections specified in this template: Introduction, Materials and Methods, Results, Discussion. Please do not reorder or retile the section headings.

The manuscript should start with a brief introduction that lays out the problem addressed by the research and describes the paper's importance. The scientific question being investigated should be described in detail. The introduction should provide sufficient background information to make the article understandable to readers in other disciplines and provide enough context to ensure that the implications of the experimental findings are clear.

Citations

Citations of references in the text should be identified using numbers in square brackets e.g., "as discussed by Cui [Cui1]" or "as discussed elsewhere [Cui1, Ninomiya1,

Li1, Wang1, Yang1]." All references should be cited within the text and uncited references will be removed.

As an example, this template includes a "sample.bib" file containing the references in BibTeX.

Citations referenced only in supplementary materials should still be included in the reference list of the main text.

Equations

Equations should be provided in a text format, rather than as an image. Equations should be numbered consecutively, in round brackets, on the right-hand side of the page by using the "`\begin{equation}`" command. They should be referred to as Equation 1, etc. in the main text.

For example, see Equation 1 and Equation 2 below.

$$a^2 + b^2 = c^2 \tag{1}$$

$$\begin{aligned} A &= \frac{\pi r^2}{2} \\ &= \frac{1}{2} \pi r^2 \end{aligned} \tag{2}$$

Upon acceptance, authors will be asked to provide the figures as separate electronic files. A figure guide is available for reference from the Science Partner Journals website. In summary, authors must supply figures as Adobe Portable Document Format (PDF), Adobe Illustrator (AI), or Encapsulated PostScript (EPS) for illustrations or diagrams; Tagged Image File Format (TIFF), JPEG, PNG, PhotoShop (PSD), EPS, or PDF for photography or microscopy. Images should be of at least 300 dpi resolution, unless due to the limited resolution of a scientific instrument. If an image has labels, the image and labels should be embedded in separate layers.

2 Methods

2.1 Literature Search Procedure

Begin with a section titled Experimental Design describing the objectives and design of the study as well as prespecified components.

2.2 Conventional Statistical Methods

2.2.1 Deletion Methods

2.2.1.1 Listwise Deletion Listwise Deletion, also known as Complete Case Analysis (CCA), is the most direct method for handling missing values. It removes an entire row from the dataset if it contains one or more missing values. The main advantage of this approach is that it is extremely simple to implement and requires no additional computation. However, this method may substantially reduce the sample size and consequently decrease statistical power.[A review on]

2.2.1.2 Pairwise Deletion Pairwise Deletion, also known as Available Case Analysis (ACA), is a method used to handle missing values. It excludes observations only when the analysis involves variables with missing data, rather than removing the entire case. The main advantage of this approach is that it retains more available data and preserves a larger sample size and statistical power. However, this method is appropriate only under the MCAR assumption. Different analyses may use different sample sizes. This can make the correlation matrix difficult to interpret and may produce a non-positive definite matrix, which can affect subsequent model estimation.

2.2.2 Mean / Median Imputation

Mean / Median Imputation is one of the most common single imputation methods in statistics. It replaces missing values with the mean or median of the observed values for that variable. This method is simple to implement and computationally efficient. It can also provide a practical solution when the dataset is very small. However, this method is appropriate only when the missing data mechanism satisfies MCAR. It may underestimate the variance and standard errors and bias the correlations among variables. In addition, the mean is sensitive to extreme values and cannot capture complex nonlinear relationships.

2.2.3 MICE

2.3 Machine Learning Methods

2.3.1 K-Nearest Neighbors (KNN)

KNN imputation is a non-parametric method because it does not assume a specific functional form or probability distribution for the data. Instead, missing values are estimated using the observed values of the K most similar instances in the dataset.[Deep learning]

This method estimates missing values using the K most similar observations in the dataset. First, the distance

between the incomplete observation and all other observations is computed using a distance metric such as Euclidean distance. Next, the K nearest neighbors are identified. For numerical variables, the missing value is imputed using the mean or weighted mean of the neighbors. If the missing value is a category, the algorithm uses the mode (the most frequent value) among the neighbors.[How To Treat]

The KNN imputation method offers several advantages over traditional missing data handling techniques such as mean imputation. First, KNN preserves the underlying relationships among variables because the imputed values are derived from observations with similar feature patterns rather than from global summary statistics. Second, KNN is a non-parametric method and therefore does not require assumptions about the underlying data distribution. This makes the method flexible and applicable to a wide range of datasets. Third, KNN can accommodate both continuous and categorical variables by using appropriate aggregation strategies such as the mean for numerical attributes and the mode for categorical attributes. Finally, because KNN relies on similarity among observations, it is capable of capturing local structures in the data, which can lead to more accurate imputations compared with simple statistical approaches.

Despite its advantages, the KNN imputation method also has several limitations. One significant issue is its computational cost. Because the algorithm must search through the entire dataset to identify the nearest neighbors for each missing value, it can become computationally expensive and memory-intensive when applied to very large datasets. Another challenge lies in its sensitivity to hyperparameters. The effectiveness of the method depends heavily on the choice of the parameter k and the quality of the features used to compute distances. If the selected features are weakly related to the variable with missing values, the imputation results may be inaccurate.[AStudyOn]

2.3.2 MissForest

MissForest is a non-parametric imputation method that utilizes Random Forest to estimate missing values. The algorithm first initializes missing entries using simple statistical methods, such as mean imputation for continuous variables and mode imputation for categorical variables. Variables are then ordered according to the amount of missing data, starting with those with the fewest missing values. For each variable with missing data, a Random Forest model is trained using the observed values as the response variable and the remaining variables as predictors. The trained model is then used to predict the missing values. This process is repeated iteratively, and the

imputed dataset is updated in each step. The algorithm stops when the difference between successive imputations begins to increase.

The main strength of MissForest is its ability to capture non-linear relationships and complex interactions between variables. Unlike methods such as KNN and MICE, MissForest can handle both continuous and categorical variables directly. It not only avoids separate processing for different data types but also eliminates the need for additional transformations, such as standardization or dummy coding. Another advantage is that it does not require strong assumptions about the data distribution. Instead, it uses random forest models to learn relationships between variables directly from the data. In addition, MissForest can estimate imputation quality using the Out-of-Bag (OOB) error, so a separate test dataset is not required.

Although MissForest has several advantages, it also has some limitations. Since the algorithm repeatedly trains random forest models during the iterative imputation process, it may require more computational time when applied to large datasets. Instead, a more relevant issue for small datasets is the potential risk of overfitting. Because MissForest relies on random forest models, the algorithm may learn patterns that are overly specific to the limited sample. Therefore, when applying MissForest to small-sample survey data, it is important to carefully examine the stability and plausibility of the imputed values.

Empirical studies suggest that MissForest performs particularly well when the dataset contains more than 1,000 observations, often achieving better imputation accuracy than traditional methods such as mean imputation or KNN-based approaches. [A review on missing] In addition, prior research indicates that for tabular datasets with moderate sample sizes (typically fewer than 30,000 observations), MissForest may outperform deep learning approaches such as GAIN or variational autoencoders (VAE). Therefore, for survey datasets with sample sizes between approximately 1,000 and 30,000 observations, MissForest can be considered a suitable method for handling missing values.[Versus]

2.4 Deep Learning Methods

The integration of deep learning into the field of data imputation primarily involves two frameworks: Generative Adversarial Networks (GANs) and Transformer-based models (Attention mechanisms).

2.4.1 Generative Frameworks

The core objective of generative frameworks is to learn the joint probability distribution of the data. Rather than sim-

ply predicting a value based on other features, these models attempt to reconstruct the underlying data-generating logic to produce imputations that maintain global statistical characteristics.

- **Generative Adversarial Networks (GANs):** These consist of a Generator, which observes available data to predict missing values, and a Discriminator, which attempts to distinguish between observed and imputed data points. GAN-based models exhibit higher resilience under high missingness rates (e.g., above 50%) because they capture the overall data generation logic rather than local correlations. GAIN [yoon2018gain] serves as a pivotal benchmark in this field, introducing a "hint vector" to guide the discriminator and force the generator to learn the true underlying distribution.
- **Variational Autoencoders (VAEs):** VAEs compress incomplete data into a low-dimensional latent space to learn the probability distribution before decoding and reconstructing the complete data matrix. A significant advantage of VAEs is their ability to provide uncertainty quantification, which is critical for risk-sensitive scenarios such as healthcare. MIWAE [mattei2019miwae] optimized importance sampling for incomplete datasets, while TVAE was specifically designed to handle tabular heterogeneity.
- **Diffusion Models:** These models simulate physical diffusion processes to reconstruct missing values through iterative denoising. TabDDPM [kotelnikov2023tabddpm] has been shown in recent 2025 literature to outperform GANs and VAEs in preserving original data distributions (e.g., minimizing KL divergence). The latest MissDDIM [zhou2025missddim] further addresses the inference latency of diffusion models by replacing stochastic sampling with a deterministic non-Markovian sampling path, achieving faster and more consistent imputation results.

2.4.2 Attention-based Frameworks

The essence of attention mechanisms lies in dynamic weight allocation. When imputing a specific missing cell, the model automatically determines which other features or similar samples are most informative for the current estimation.

- **TabNet:** This architecture utilizes sequential attention to "mask" features at each decision step, mimicking the splitting logic of decision trees. By employing the Sparsemax function to generate sparse

weights, TabNet [arik2021tabnet] achieves tree-like interpretability while effectively modeling complex non-linear interactions between features.

- **Transformer Variants:** Through self-attention mechanisms, these models capture not only "between-feature" correlations within a single row but also "between-sample" relationships across different rows. FT-Transformer [gorishniy2021revisiting] transforms features into embeddings for processing, while the Non-Parametric Transformer (NPT) [kossen2021self] allows the model to communicate across all samples in a dataset. Recent research has further optimized these frameworks to enhance fault tolerance and computational efficiency.
- **Tabular Foundation Models and In-Context Learning (ICL):** Frameworks such as TabPFN eliminate the need for retraining on missing data. Instead, they treat the training set as "context" for the Transformer, leveraging pre-trained weights to predict missing values in the test set via a single forward pass. TabImpute [feitelberg2025tabimpute] introduced entry-wise featurization, achieving a 100x speedup over traditional TabPFN imputation methods and setting a new benchmark for zero-shot data imputation.

3 Results

The results should describe the experiments performed and the findings observed. The results section should be divided into subsections to delineate different experimental themes.

- All data should be presented in the Results. No data should be presented for the first time in the Discussion. Data (such as from Western blots) should be appropriately quantified.
- Subheadings must be either all complete sentences or all phrases. They should be brief, ideally less than 10 words. Subheadings should not end in a period. Your paper may have as many subheadings as are necessary.
- Figures and tables must be called out in numerical order. For example, the first mention of any panel of Fig. 3 cannot precede the first mention of all panels of Fig. 2. The supplementary figures (for example, fig. S1) and tables (table S1) must also be called out in numerical order.

4 Discussion

In this section, we will introduce the experiments conducted to evaluate the differences between traditional machine learning and deep learning methods for data imputation. We designed a two-phase experimental approach: Phase 1 compares the data recovery accuracy of different imputation methods, and Phase 2 evaluates the prediction accuracy for gender classification after imputation.

4.1 Experimental Setup

4.1.1 Environment and Tools

All experiments were conducted on the Kaggle online platform to ensure a consistent computing environment, with a fixed random seed of 42 applied to all data splitting and model initialization processes to ensure reproducibility.

4.1.2 Dataset Descriptions

To evaluate the impact of sample size on different imputation techniques, we selected two datasets with significantly different scales:

- **Small-scale Dataset (Boy or Girl):** This dataset contains approximately 400 samples, representing a typical small-sample scenario. It includes various features such as height, weight, star sign, phone OS, and IQ. The target variable is gender, which is used to evaluate the performance of imputation methods on downstream classification tasks.
- **Large-scale Dataset (CDC BRFSS):** Downloaded via the kagglehub library, this dataset contains hundreds of thousands of health-related records from the Behavioral Risk Factor Surveillance System (BRFSS). This large-scale dataset is used to assess whether deep learning models demonstrate superior representation capabilities when a vast amount of data is available.

4.1.3 Experimental Design and Protocol

We use a control-variable approach to ensure that all processes remain identical except for the choice of the imputation method. The experimental pipeline is organized as follows:

- **Data Preprocessing and Cleaning:** The raw datasets undergo initial cleaning to handle anomalies. Categorical features are transformed using *Label Encoding* to ensure they are compatible with both machine learning and deep learning algorithms.

- **Phase 1: Recovery Accuracy Analysis:** To test the reconstructive capability of each method, we simulate data loss by randomly masking 10% of the values in the continuous features. Various imputation techniques are then applied to fill these gaps, and their performance is evaluated using distance-based metrics (such as MAE and R^2) by comparing the imputed values against the original ground truth.
- **Phase 2: Downstream Performance Evaluation:** This phase investigates the impact of different imputation methods on the final predictive outcome. Using a unified downstream classification model, we measure the gender prediction accuracy. This allows us to determine which imputation strategy provides the highest utility for practical classification tasks.

4.2 Phase 1: Recovery Accuracy Analysis

4.2.1 Experiment Objectives and Method Selection

In this experiment, we picked four methods based on the analysis from Chapters 2 and 3. We chose MICE and MissForest to represent traditional machine learning, and GAIN and VAE for deep learning. By comparing these four, we want to observe two main things: first, which method is the most accurate at restoring missing values under the same conditions; second, how sample size affects performance. Basically, we want to find out if deep learning is superior for large datasets, and if traditional machine learning is actually more accurate when we only have a small sample size.

4.2.2 Experimental Design and Evaluation Metrics

The experimental design focuses on two datasets: the *Boy or Girl* dataset and the *CDC BRFSS* (detailed in Section 4.1.2). To evaluate the reconstructive performance of different imputation models on continuous data, we specifically target the "Height" feature for our experiments. We simulate a missing data scenario by randomly masking 10% of the observations, using the remaining 90% as the training baseline for the models.

To quantify the effectiveness of the imputation as a regression task, we utilize four metrics to assess the fit between the imputed values and the ground truth:

First, **Mean Absolute Error (MAE)** measures the average absolute difference between the imputed values and the actual values, providing a direct reflection of the

prediction error scale:

$$MAE = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i| \quad (3)$$

Second, **Root Mean Square Error (RMSE)** is employed to capture the stability of the imputation. Compared to MAE, RMSE is more sensitive to large errors due to the squaring of differences, which helps identify potential outliers in the results:

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2} \quad (4)$$

Third, the **Coefficient of Determination (R^2)** is used to evaluate the model's ability to explain the variance in the original data. An R^2 value closer to 1 indicates that the imputed data closely follows the distribution of the original dataset:

$$R^2 = 1 - \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{\sum_{i=1}^n (y_i - \bar{y})^2} \quad (5)$$

Finally, we include **Bias** to observe if the model exhibits systematic errors, such as a tendency to consistently over- or under-estimate the height values:

$$Bias = \frac{1}{n} \sum_{i=1}^n (\hat{y}_i - y_i) \quad (6)$$

By analyzing these four metrics collectively, we can comprehensively evaluate each imputation method's performance across different data scales in terms of accuracy, stability, and distribution fitting.

4.2.3 Recovery Performance on Small Dataset

In the tests conducted on the small-scale "Boy or Girl" dataset, the experimental results align with the initial hypothesis, demonstrating that traditional machine learning significantly outperforms deep learning models in terms of data recovery capability. According to the observations in Table 1, MissForest stands out across all metrics, achieving the highest R^2 of 0.2131 and the lowest MAE of 3.9075. This confirms that the random forest-based imputation method, even with limited data, can accurately capture the non-linear correlations between height and other features through its ensemble learning mechanism.

In contrast, the performance of the two deep learning models, GAIN and VAE, is highly unsatisfactory. Their MAE values are as high as 8.3544 and 5.8562, respectively, which are not only considerably higher than those of MissForest but also behind the control group, Median Impu-

tation ($MAE = 4.9310$). More significantly, both models exhibit negative R^2 values, implying that generative models completely fail to capture the data distribution in an environment with approximately 400 samples, with their prediction biases exceeding the statistical baseline.

This phenomenon validates the perspective presented in Chapter 3: deep learning models rely heavily on large volumes of samples for weight optimization and feature representation learning. In scenarios involving simple feature dimensions and minute sample sizes, neural networks are prone to getting stuck in local optima or suffering from severe overfitting, which prevents the models from converging effectively.

Table 1: Recovery Performance Comparison - Small Dataset

Method	MAE (cm)	RMSE (cm)	R^2	Bias (cm)
Median Impute	4.9310	5.8398	-0.1278	-1.9655
MICE	4.5183	6.5659	0.1341	0.8883
MissForest	3.9075	4.8780	0.2131	-1.6122
GAIN	8.3544	9.2752	-0.7280	-6.0957
VAE	5.8562	7.4658	-0.1196	-0.5622

Following the numerical evaluation in Table 1, we examine the visual distribution. The following figures show the relationship between actual values and imputed values for each method.

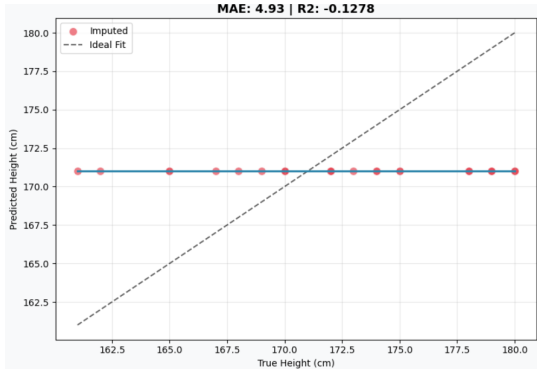


Figure 1: Median Imputation (Baseline). It shows a horizontal line as it fills all gaps with a median value.

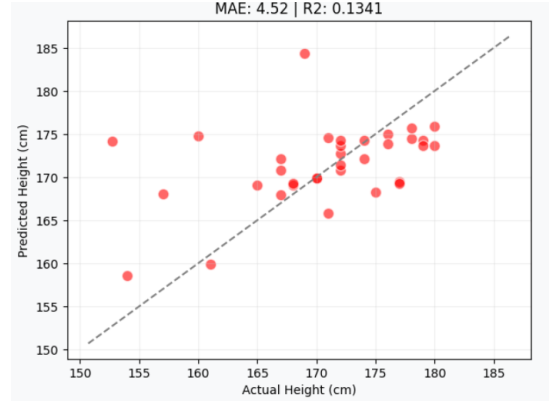


Figure 2: MICE Imputation. Shows some correlation but with significant dispersion.

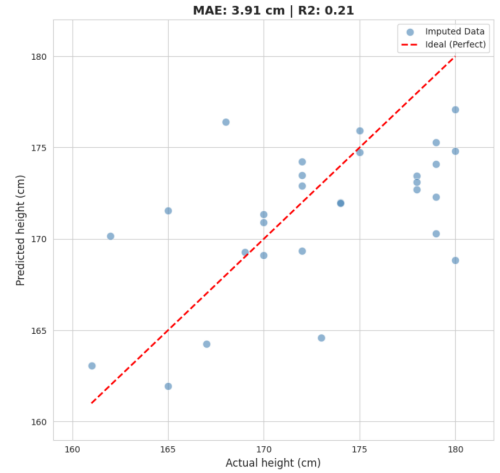


Figure 3: MissForest Imputation. This method shows the highest accuracy with points closest to the diagonal.

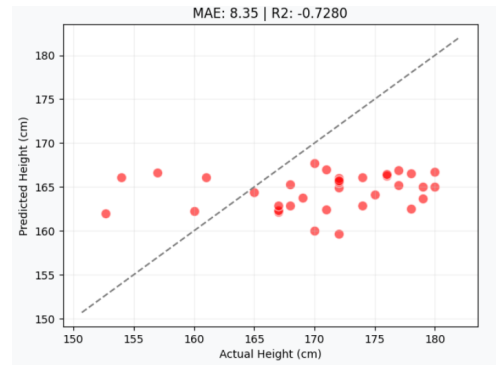


Figure 4: GAIN. The points are widely scattered, indicating poor convergence on small datasets.

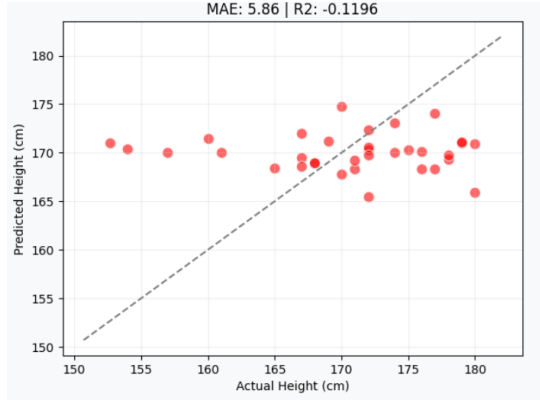


Figure 5: VAE. Like GAIN, it fails to capture the linear trend in this small-sample scenario.

In summary, for small-scale datasets with simple feature dimensions, deep learning models are not the optimal choice for missing value imputation, as their recovery performance often lags behind basic statistical baselines (such as Median Imputation). Conversely, the ensemble learning mechanism in traditional machine learning (e.g., MissForest) demonstrates superior robustness and effectively captures feature correlations even with limited data. These findings suggest that when handling tabular data, model selection must prioritize the impact of data scale on algorithmic convergence; pursuing complex models without sufficient data may lead to lower imputation fidelity.

4.2.4 Recovery Performance on Big Dataset

In the tests using the large-scale dataset (CDC BRFSS), the experimental results show a completely different trend compared to the small-sample tests. When the amount of data increases significantly, the advantages of deep learning models start to appear. According to the data in Table 2, VAE showed a dominant advantage in all evaluation metrics. Its R^2 reached the highest in the entire group at 0.9386, and the MAE was only 0.724. This shows that in a big data environment, VAE can accurately learn the data distribution and the complex relationships between features.

In contrast, traditional machine learning methods like MICE and MissForest still maintain good stability, with R^2 values of 0.5572 and 0.5664, respectively. However, their recovery accuracy is clearly behind VAE. This proves that while traditional methods can still provide reliable results when dealing with massive amounts of data, their ability to represent data has reached a bottleneck. They find it hard to improve the imputation quality further through big data, unlike certain deep learning models.

It is worth noting that another deep learning model, GAIN, still performs poorly at this stage. The failure

of GAIN is mainly due to the gap between distribution matching and point-wise accuracy. While GAIN focuses on making the overall statistical distribution look real to fool the discriminator, it often ignores the exact precision for each individual data point, leading to a low R^2 . Furthermore, GAIN often suffers from "mode collapse," where the generator simply predicts the average height to stay "safe," which explains the horizontal patterns seen in its scatter plots. Additionally, the adversarial training process is highly unstable for tabular data, making it difficult to converge even with large datasets. In summary, the large-scale experiment confirms that when the data scale is large enough, deep learning—especially generative models like VAE—has great potential to surpass traditional methods, provided the training is stable.

Table 2: Recovery Performance Comparison - Big Dataset

Method	MAE (cm)	RMSE (cm)	R^2	Bias (cm)
Median Impute	8.8916	10.8688	-0.0927	-1.9041
MICE	5.4667	7.0170	0.5572	0.0222
MissForest	5.4062	6.9443	0.5664	0.0045
GAIN	8.118	10.0684	0.0791	-2.4617
VAE	0.724	2.576	0.9386	0.0664

In this section, we visually compare the imputation results for the CDC BRFSS dataset. The following figures demonstrate how each model performs as the sample size increases.

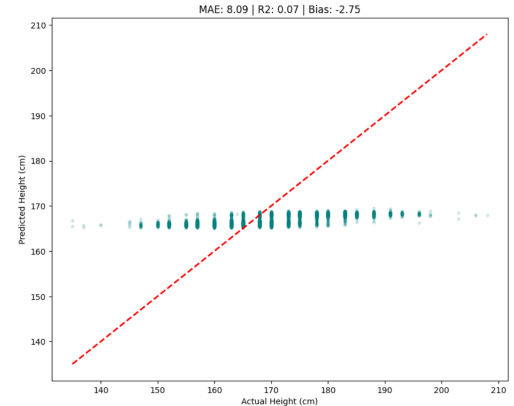


Figure 6: Median Imputation. Similar to the small dataset, it fails to capture any individual variance.

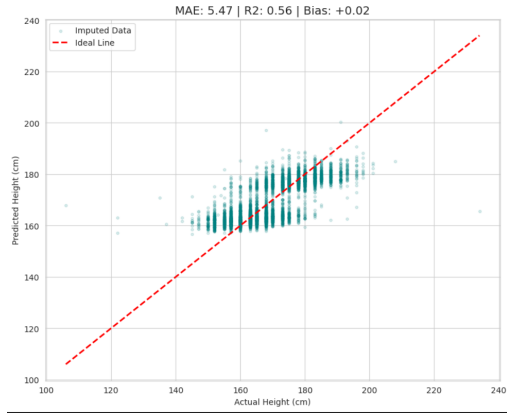


Figure 7: MICE Imputation. The distribution begins to align with the diagonal, but still shows noticeable spread.

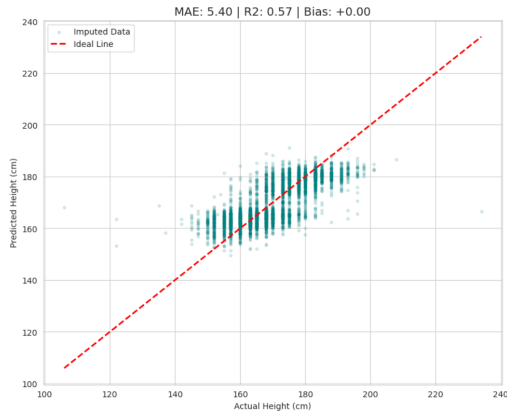


Figure 8: MissForest Imputation. Performance is stable, but the density around the diagonal is lower than VAE.

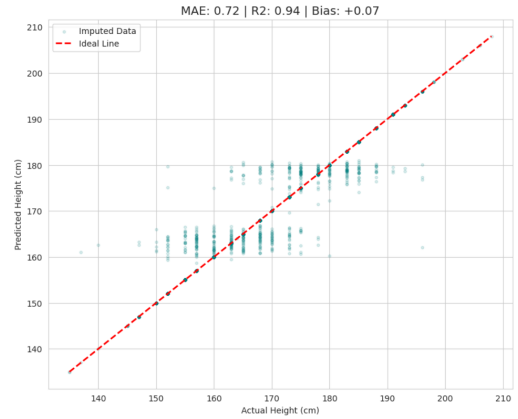


Figure 10: VAE. The points are highly concentrated along the $y = x$ line, visually confirming the $R^2 = 0.9386$ result.

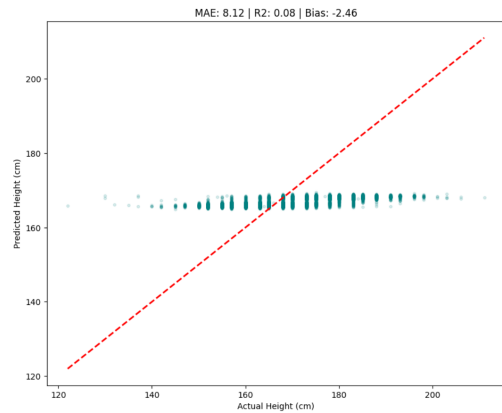


Figure 9: GAIN. The points are widely scattered and fail to align with the diagonal line, which visually explains the very low R^2 of 0.0791 and high MAE.

In our large-scale experiment, we observed a clear reversal in performance. As the data size grows, the recovery accuracy of traditional machine learning remains stable, but it seems to have a limit. On the other hand, with enough data support, certain deep learning methods show a strong ability to learn features, which greatly improves the imputation accuracy. This proves that for handling missing values in large datasets, generative deep learning models are a better and more promising solution.

Funding

Name financially supporting bodies (written out in full), followed by the funding awardee and associated grant numbers (if applicable) in square brackets.

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“This work was supported by the Engineering and Physical Sciences Research Council [grant numbers xxxx, yyyy]; the National Science Foundation [grant number zzzz]; and a Leverhulme Trust Research Project Grant.”

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